Throughout the 1700s, scientists (then called natural philosophers) believed that gravity was governed by Newton's laws, and that light was made of corpuscles (particles) that are emitted by their sources at a very high, universal speed. That speed was known to be about 300,000 kilometers per second, thanks to telescopic measurements of light coming from Jupiter's moons as they orbit around their parent planet.

In 1783 John Michell, a British natural philosopher, dared to combine the corpuscular description of light with Newton's gravitation laws and thereby predict what very compact stars should look like. He did this by a thought experiment which I repeat here in modified form:

Launch a particle from the surface of a star with some initial speed, and let it move freely upward. If the initial speed is too low, the star's gravity will slow the particle to a halt and then pull it back to the star's surface. If the initial speed is high enough, gravity will slow the particle but not stop it; the particle will manage to escape. The dividing line, the minimum initial speed for escape, is called the "escape velocity." For a particle ejected from the Earth's surface, the escape velocity is 11 kilometers per second; for a particle ejected from the Sun's surface, it is 617 kilometers per second? or 0.2 percent of the speed of light.

Michell could compute the escape velocity using Newton's laws of gravity, and could show that it is proportional to the square root of the star's mass divided by its circumference. Thus, for a star of fixed mass, the smaller the circumference, the larger the escape velocity. The reason is simple: The smaller the circumference, the closer the star's surface is to its center, and thus the stronger is gravity at its surface, and the harder the particle has to work to escape the star's gravitational pull.

There is a *critical circumference*, Michell reasoned, for which the escape velocity is the speed of light. If corpuscles of light are affected by gravity in the same manner as other kinds of particles, then light can barely escape from a star that has this critical circumference. For a star a bit smaller, light cannot escape at all. When a corpuscle of light is launched from such a star with the standard light velocity of 299,792 kilometers per second, it will fly upward at first, then slow to a halt and fall back to the star's surface; see Figure 3.1.